

KAMP: Knowledge-Anchored Multimodal Pretraining for Medical Image

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Abstract

Cross-modal biomedical signals such as pathology and genomics can provide richer and more robust semantic guidance for medical image pretraining. However, semantic guidance remains insufficient because privacy constraints severely limit paired imaging and biomedical data. A further challenge is modality discrepancy, which propagates intra-modal statistical bias and cross-modal noise and degrades representation quality. To this end, we propose KAMP, an LLM-driven, multimodal biomedical-guided pretraining framework for medical images. KAMP derived textual priors and uses them as semantic anchors to align medical images with multimodal biomedical data, yielding rich and robust medical image representations under scarce paired data. Specifically, KAMP consists of three stages: first, the LLM conditions on imaging metadata and clinical text to generate personalized, multi-target knowledge that the contrastive aligner uses as semantic anchors to align medical images with multimodal biomedical data; second, we optimize the LLM with GRPO using the previously trained aligner as the reward model; third, we re-train the aligner using the optimized knowledge as semantic anchors, further strengthening image representations while suppressing bias and noise introduced by the other modalities. Comprehensive evaluations across brain, bladder, and liver cancer datasets demonstrate KAMP's superiority on downstream few-shot prediction.

1. Introduction

Multimodal biomedical data[4, 11, 15, 15, 31, 54] provides rich and comprehensive semantics, enabling more faithful understanding and better task performance. Pathology, genomics, and imaging each reveal distinct biological signals: pathology provides tissue patterns and cellular context, excelling at histologic subtyping and grading; genomics reveals the underlying molecular mechanisms, offering insights into tumor behavior and biological heterogeneity; Medical image provides organ-level views that localize lesions and quantify disease burden. However, privacy con-

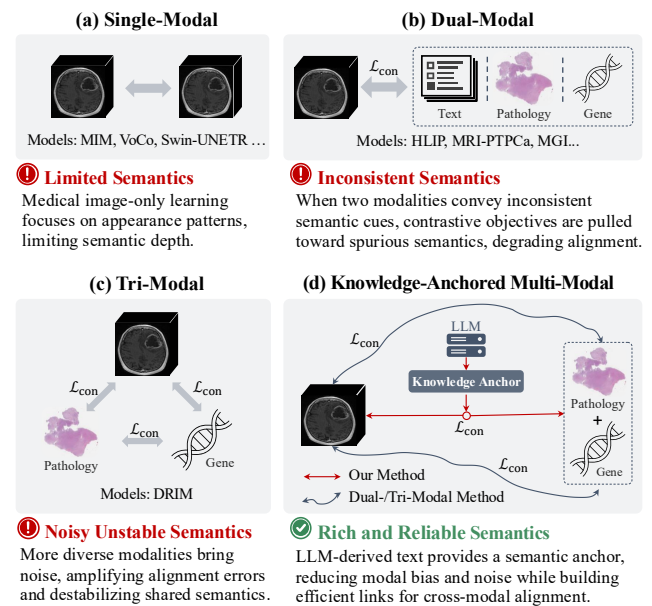


Figure 1. Comparison of existing medical image pretraining paradigms and the proposed KAMP. (a) Single-modal learning trains on medical images only. (b) Dual-modal learning aligns medical images with text, pathology, or genomics via contrastive objectives. (c) Tri-modal contrastive learning jointly aligns medical images with pathology and genomics. (d) KAMP performs knowledge-anchored cross-modal contrastive learning that aligns medical images with pathology and genomics, using LLM-derived text as a semantic anchor that stabilizes and improves alignment.

straints and clinical workflows mean that pathology and genomics require invasive procedures and are available for far fewer patients, while medical images are relatively easier to acquire. These constraints motivate pretraining medical imaging models with rich semantics that transfer to downstream tasks, enabling accurate diagnosis when pathology and genomics are scarce, reducing reliance on invasive biopsies and costly genomic assays.

However, prevailing medical image pretraining remains hampered by semantic insufficiency and modality bias. As shown in Fig. 1(a), single modality learning [9, 39, 41, 46,

55, 57] optimizes image-only objectives such as contrastive discrimination, spatial context prediction, and masked reconstruction, which emphasize superficial appearance cues, curtail semantic expressivity and constrain semantic depth. As shown in Fig. 1(b), Dual-modality learning [1, 10, 18–20, 50, 52] aligns medical images with a companion source such as text reports, pathology, or genomics. When inconsistencies arise between the paired sources, the contradictory signals steer the contrastive objective toward spurious correspondences, driving the alignment toward an erroneous embedding space and degrading downstream performance, especially when paired training data are limited. As shown in Fig. 1(c), Tri-modality learning [31] aligns medical imaging simultaneously with pathology and genomics, by introducing one more modality, which supplies extra evidence that partly alleviates pairwise inconsistencies, yet the increased modality diversity also injects noise and yields error-amplified, unstable alignment.

Large language models (LLMs) [27, 49] offer a principled way to densify supervision with domain priors, mitigating semantic insufficiency and modality bias, but current LLM applications ignore the imaging setting (*e.g.*, MRI sequence or CT phase) and patient-specific context, which often causes generated text descriptions to drift from the actual visual evidence. Concretely, even under the same diagnosis label, patient-specific phenotypes (*e.g.*, age, gender, tumor grade, prior treatment) induce substantial appearance variability of medical images. When text generation is not conditioned per patient, LLMs average over these cues and yield template-like text that is weakly grounded in the medical image, degrading text–image alignment; prior work [16, 17, 30] likewise omits personalized conditioning. Moreover, prior approaches [16, 17, 30] expand labels into richer descriptions, but typically condition only on the imaging type (CT or MRI) rather than the imaging setting that determines what the medical image can show. Different imaging settings produce large visual differences; for example, in the T1-weighted contrast-enhanced MRI (T1C), blood vessels are bright and well delineated, whereas in the T1-weighted pre-contrast MRI (T1W), the same vessels are faint or even invisible. Consequently, LLM-generated vessel descriptions conditioned only on the imaging type may fit T1C but contradict T1W, resulting in a mismatch between text and image that corrupts supervision and skews the learned alignment.

To address the aforementioned challenges, we introduce KAMP, a new multimodal pretraining framework for medical images. As shown in Fig. 1(d), the core idea of KAMP is to leverage LLM-derived textual knowledge as a semantic anchor, facilitating efficient alignment between medical images and multimodal biomedical data, enabling complementary information from multimodal biomedical data to guide medical image representation learning in a semanti-

cally enriched manner. KAMP operates in three stages: (1) We instantiate a cross-modal aligner with two branches: a medical image branch and a multimodal biomedical branch. A frozen LLM, conditioned on the imaging setting in the metadata and the patient’s clinical text, generates personalized, visibility-aware descriptions. We inject these texts into the medical image branch, while the biomedical branch fuses raw clinical text, pathology, and genomics into a compact embedding; a symmetric contrastive objective between the two branches then trains the initial aligner. (2) We freeze the aligner and optimize the LLM with GRPO, using the aligner as a reward model that scores each generated text by the similarity it induces between medical image branch and multimodal biomedical branch embeddings. The policy is updated to favor knowledge that yields higher alignment scores, implicitly steering the LLM toward personalized descriptions that are visible under the given imaging setting. (3) We freeze the LLM and retrain the aligner using knowledge sampled from the optimized policy, injecting the refined texts into the aligner and optimizing the same symmetric contrastive objective. This stage enables the aligner to incorporate the optimized knowledge into its representation space, strengthening semantic consistency and mitigating modality-induced bias and noise. Our contributions are fourfold:

- (1) We propose KAMP, a knowledge-anchored multimodal pretraining paradigm for medical images, which achieves strong performance in few-shot regimes;
- (2) KAMP employs a reinforcement learning strategy GRPO [35] to tune the LLM [49] conditioned on patient context information, deriving personalized visibility-aware priors that strengthen medical image representations;
- (3) KAMP leverages textual knowledge as an anchor to build efficient links for cross-modal semantic alignment, achieving modality-bias-robust information guidance;
- (4) We demonstrate SOTA few-shot results across three downstream task families (cancer subtyping, grading, and microvascular invasion prediction), covering multiple cancer types.

2. Related Work

Pretraining Paradigms for Medical Images. Recent medical image pretraining has largely advanced along three lines that motivate our design. First, single-modality pretraining relies on image-only signals, such as contrastive learning [39, 41, 57], spatial position prediction [46], reconstruction [9, 55], and supervised transfer pretraining [5], delivering steady gains while offering limited semantic grounding. Second, dual-modality alignment seeks to couple medical images with text, genomics, or pathology. Medical image-text report pretraining [1, 10, 18–20, 50–52] drives progress on retrieval, recognition, report generation, but report supervision is coarse. Medical image-gene align-

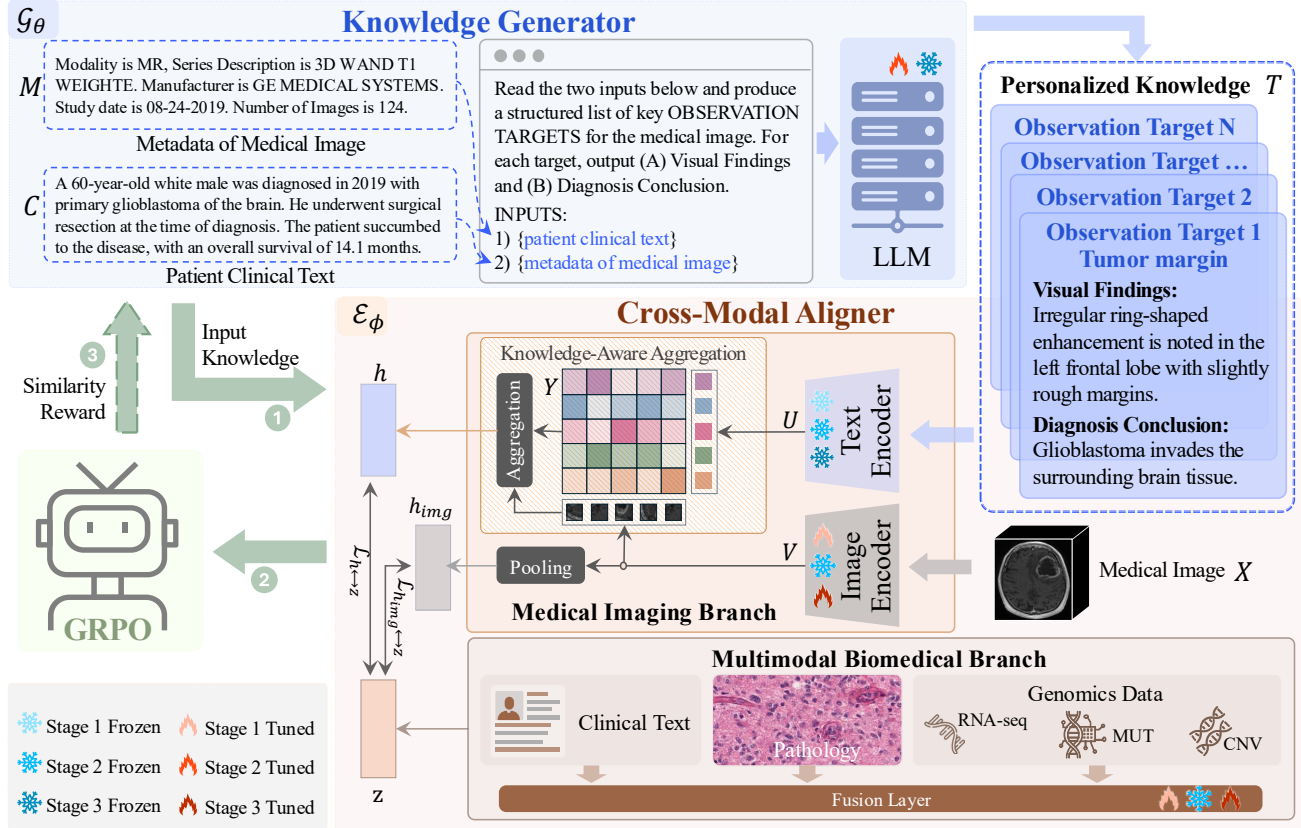


Figure 2. Overview of the proposed KAMP framework. The system comprises two components: (a) *Knowledge Generator*, which conditions on imaging metadata and patient clinical text to produce personalized multi-target knowledge; (b) *Cross-Modal Aligner*, which includes a *Medical Image Branch* that aggregates image tokens with the generated knowledge to produce a knowledge-aggregated image embedding and also yields an image-only embedding, and a *Multimodal Biomedical Branch* that fuses clinical text, whole slide pathology, and genomics into a biomedical embedding. Alignment is learned with a symmetric contrastive objective between the knowledge-aggregated image embedding and the biomedical embedding, complemented by a contrastive term between the image-only embedding and the biomedical embedding. Training follows a three-stage schedule with teacher-guided pre-alignment, aligner as reward GRPO refinement of the generator, and a final alignment using the refined knowledge.

ment [23, 38, 56] demonstrates that pairing images with genomics improves representation learning, yet the large gap in semantic granularity renders alignment indirect, hindering robust alignment. Medical image-pathology alignment [33] has shown strong results by coupling medical image with whole slide histology, but lacking genomics, it can't access the deeper molecular drivers that make the disease complex. Third, tri-modality alignment of radiology, pathology, and genomics is nascent; DRIM [31] equips the encoder with a tri-modal alignment module, but an expanded modality set accumulates noise, preventing stable high-level alignment. To address these limitation, our framework treats LLM-derived knowledge as an semantic anchor that is used to guide cross-modality alignment. This design densifies medical-image semantics without requiring fully paired data at scale and mitigates cross-modal noisy and interference by rewarding knowledge that tight-

ens alignment.

LLM-enhanced Medical Vision. Large language models (LLMs) are increasingly used to densify supervision for medical vision encoders. For radiology, BiomedCoOp, MedUnA, and UniMed-CLIP employ LLM-generated descriptions to expand labels and strengthen image-text alignment [16, 17, 30]. These methods are typically conditioned on label and thus provide corpus-wide, generic guidance; they do not systematically incorporate the imaging setting or patient-specific context that governs visibility and often decides whether a cue is present, which can make generated text deviate from the actual image and limit its value for representation learning. Our framework instead conditions on metadata and clinical text to generate personalized multi-target knowledge that is faithfully grounded to the visual evidence, reducing text-image drift and providing a robust, low-bias semantics. Several works [8, 28, 36] apply

LLM guidance to computational pathology, steering pathology feature learning, yet pathology acquisition is invasive and available for fewer patients, which constrains practical reach scale.

Biomedical multimodal fusion. Transformer architectures [7, 25, 44] have become the standard method for multimodal fusion, where tokenized modalities interact through attention to exchange information while preserving modality specific structure. For medical image and text fusion, GLORIA [14], MedKLIP [45], and MG-3D [26] couple visual and textual tokens with cross attention, enabling fine-grained fusion in radiology. For pathology and genomics, HEALNet [11], SURVPATH [15], MMP [37], MOTCat [48], CMTA [54], and MCAT [4] use transformer-based fusion to combine slide-level morphology with genomics while handling heterogeneity, scale disparity, and missing modalities. Guided by these developments, our medical image branch employs the cross-attention method to aggregate visual tokens with textual knowledge, and our biomedical branch fuses clinical text, pathology, and genomics inspired by HEALNet, yielding a compact and robust fusion embedding.

3. Method

The overall KAMP framework is illustrated in Fig. 2. and comprises two core components: a knowledge generator and a cross-modal aligner. The knowledge generator produces initial knowledge candidates using GPT5 [27] and subsequently iteratively refines these candidates through reinforcement learning based on Qwen3-8b [49]. The cross-modal aligner contains two branches, a medical image branch and a multimodal biomedical branch, each of which learns representations via contrastive learning. In Sec. 3.1, we motivate the choice of generated knowledge as anchors for contrastive learning. In Sec. 3.2, we detail the reinforcement learning framework for knowledge generation. In Sec. 3.3, we detail the design of the knowledge-anchored contrastive learning.

3.1. Knowledge Anchor

Let h denote the medical image representation, z the multimodal biomedical evidence, and A the LLM-derived personalized knowledge, which serves as the anchor. The chain rule of mutual information[6] gives:

$$I(h; z, A) = I(h; z) + I(h; A | z). \quad (1)$$

Since conditional mutual information is nonnegative, $I(h; A | z) \geq 0$, we have $I(h; z, A) \geq I(h; z)$. Writing mutual information with entropies yields

$$I(h; z) = H(h) - H(h | z), \quad (2)$$

$$I(h; z, A) = H(h) - H(h | z, A), \quad (3)$$

and thus

$$H(h | z, A) = H(h | z) - I(h; A | z). \quad (4)$$

Eq. (4) shows that conditioning on the anchor can only decrease the residual uncertainty of h given z , because $H(h | z, A) \leq H(h | z)$, and it quantifies the exact reduction as

$$\Delta H \triangleq H(h | z) - H(h | z, A) = I(h; A | z) \geq 0. \quad (5)$$

Intuitively, the anchor A shares information with both the image h and the context z ; it complements the portion of h that z alone cannot explain, therefore conditioning on A after z removes leftover uncertainty. The removed amount is exactly the conditional mutual information in Eq. (5), that is $\Delta H = I(h; A | z) \geq 0$, which measures how much additional image information is still shared with the anchor under the known context z . If A is conditionally independent of h given z , then $I(h; A | z) = 0$ and conditioning on A does not further reduce uncertainty. In our setting the anchor is personalized knowledge mined by the LLM, so it captures rich imaging cues that are predictive yet often missing from z or inconsistently reflected across modalities; as a result $I(h; A | z) > 0$ in practice and the entropy reduction in Eq. (5) is positive. Conditioning on A projects h and z onto a common latent space, pruning spurious cross-modal correlations and reducing modality bias, which improves alignment.

3.2. LLM-Derived Knowledge Generation

Stage 1 Teacher-Knowledge Align. In stage 1, we employ GPT5[27] as a teacher generator to produce textual knowledge, which is integrated into the cross-modal aligner to enrich medical image semantics and improve alignment.

Conditioned on medical image metadata M (full input, primarily its imaging setting), a teacher generator GPT5[27] produces a visibility set that enumerates clinically salient visual cues emphasized by the setting and relevant to tumor assessment; we refer to each cue as a target. Next step, conditioned on targets and patient clinical text C , a teacher generator produces patient-specific descriptions for each target consisting of two fields, a visual finding text and a diagnostic conclusion. The teacher knowledge $T_{teacher}$ contains P target-wise texts restricted to targets visible under the present imaging setting and individualized using C . This design yields precise semantic signals that respect modality-dependent visibility while preserving patient specificity. (see supplementary material).

The teacher knowledge is fed to a cross-modal aligner $\mathcal{E}_\phi$, a two-branch architecture that aligns representations from a medical image branch and a multimodal biomedical branch. (1) On the medical image branch, we integrate teacher knowledge into the 3D image features to obtain an enhanced representation. Given a 3D medical image

$X \in \mathbb{R}^{C \times Z \times H \times W}$, the image encoder partitions X and outputs token features $V \in \mathbb{R}^{S \times d}$, where S is the number of volume tokens. Each medical image is paired with teacher knowledge $T_{teacher}$ containing P target texts; the text encoder maps $T_{teacher}$ to $U \in \mathbb{R}^{P \times d}$. The cross-attention-based aggregation of U and V produce an embedding $h \in \mathbb{R}^d$. In addition, before cross attention we apply token pooling to V to obtain an image-only embedding $h_{img} \in \mathbb{R}^d$. (2) On the multimodal biomedical branch, we encode patient’s raw clinical text, whole-slide pathology image, and gene profiles from three tabular sources, and fuse them with a fusion layer[12] to obtain the multimodal representation $z \in \mathbb{R}^d$.

For a batch of size B and a learnable temperature $\tau > 0$, the loss between h and z is

$$\mathcal{L}_{h \leftrightarrow z} = -\frac{1}{2B} \sum_{b=1}^B \log \frac{\exp(h_b^\top z_b / \tau)}{\sum_{b'=1}^B \exp(h_b^\top z_{b'} / \tau)} - \frac{1}{2B} \sum_{b'=1}^B \log \frac{\exp(z_{b'}^\top h_{b'} / \tau)}{\sum_{b=1}^B \exp(z_{b'}^\top h_b / \tau)}. \quad (6)$$

So the aligner objective is

$$\mathcal{L}_{align} = \mathcal{L}_{h \leftrightarrow z} + \mathcal{L}_{h_{img} \leftrightarrow z}. \quad (7)$$

The first stage aims to establish an initial cross-modal alignment using teacher knowledge.

Stage 2 Aligner-as-Reward RL with GRPO. We freeze the Stage-1 cross-modal aligner $\mathcal{E}_\phi$ and use it as a reward model to optimize the knowledge generator Qwen3-8B [49] $\mathcal{G}_\theta$ with Group Relative Policy Optimization (GRPO) [35].

At the start of each GRPO iteration, we snapshot the knowledge generator (Qwen3-8B) to obtain a frozen behavior policy $\pi_{\theta_{old}}$, from which we sample G candidates $\{T^g\}_{g=1}^G$ conditioned on (M, C) . A trainable copy with the same initialization, denoted π_θ , is then updated on these samples using the GRPO objective.

For each T^g , the frozen medical image branch of $\mathcal{E}_\phi$ integrates the text with 3D image features to produce the knowledge-aggregated medical image embedding h^g ; in parallel, the frozen biomedical branch provides z from the patient’s other biomedical modalities. The alignment reward is the cosine similarity $s(\cdot, \cdot)$:

$$r^g = s(h^g, z). \quad (8)$$

We compute group-relative advantages by normalizing rewards within the group: $A^g = \frac{r^g - \bar{r}}{\sigma}$, where $\bar{r}$ and σ are the group mean and sample standard deviation.

The GRPO objective with PPO-style clipping and an ex-

plicit KL regularizer to a reference policy π_{ref} is

$$\mathcal{L}_{GRPO} = -\mathbb{E} \left[\sum_{g=1}^G \min \left(\frac{\pi_\theta(T^g | M, C)}{\pi_{\theta_{old}}(T^g | M, C)} A^g, \text{clip} \left(\frac{\pi_\theta(T^g | M, C)}{\pi_{\theta_{old}}(T^g | M, C)}, 1-\epsilon, 1+\epsilon \right) A^g \right) \right] + \beta \mathbb{E} [D_{KL}(\pi_\theta(\cdot | M, C) \| \pi_{ref}(\cdot | M, C))]. \quad (9)$$

We update the LoRA adapters of $\mathcal{G}_\theta$, keep $\mathcal{E}_\phi$ frozen, and refresh π_{ref} at the start of each outer iteration as in GRPO. Intuitively, the aligner “rewards” candidates that drive the medical image embedding toward the patient’s multimodal biomedical representation.

Stage 3 GRPO-Enhanced Align. We freeze the knowledge generator $\mathcal{G}_\theta$, sample knowledge $T \sim \pi_\theta(\cdot | M, C)$, and feed T to the aligner $\mathcal{E}_\phi$ as in Stage 1, aggregating text with 3D image features and optimizing $\mathcal{E}_\phi$ with the symmetric two-branch contrastive objective between (h, h_{img}) and z . This adaptation exposes the aligner to the knowledge distribution induced by the policy optimized in Stage 2, allowing its representations to adjust to personalized multi-target knowledge produced by π_θ , thus sharpening semantic representations and tightening alignment.

3.3. Prior Knowledge Injection

We introduce the knowledge anchor A (see Sec. 3.1) within the two-branch aligner $\mathcal{E}_\phi$, casting LLM-derived textual knowledge as a semantic reference across branches to reduce cross-modal bias, stabilize alignment. Specifically, given a 3D medical image partitioned into S volumes, the image encoder outputs $V \in \mathbb{R}^{S \times d}$. For each medical image we have P LLM-generated target descriptions; the text encoder yields $U \in \mathbb{R}^{P \times d}$. We compute the cross-attention weights

$$Y = \text{softmax} \left(\frac{UW_Q(VW_K)^\top}{\sqrt{d}} \right) \in \mathbb{R}^{P \times S}, \quad (10)$$

where $W_Q, W_K \in \mathbb{R}^{d \times d}$. Row p of Y contains the weights over the S volumes for target p . We then aggregate these weights with the medical image feature to obtain text-aggregated visual responses $h \in \mathbb{R}^d$. Incorporating the target descriptions in this way enables fine-grained localization of target-relevant structures and injects LLM-derived domain priors, enriching the resulting medical-image semantics.

The LLM-derived priors guide cross attention over image tokens; the resulting responses form a compact knowledge anchor A that is embedded in the medical image representation h . As shown in Sec. 3.1, given the biomedical context z , further conditioning on the knowledge anchor A cannot increase the residual uncertainty of h , and

Table 1. Few-shot classification performance (presented in %) on three datasets. Best results are in bold and second-best are underlined.

Datasets	Models	3-shot	5-shot	10-shot	15-shot	20-shot
BraTS23-MEN (3 classes)	DRIM	<u>60.2±5.1</u>	54.7±13.6	57.1±10.7	56.7±8.6	60.9±7.9
	MRI-PTPCa	45.6±5.9	48.7±8.3	58.0±4.7	57.1±4.0	58.4±3.7
	HLIP	57.9±8.7	<u>58.1±8.8</u>	<u>60.8±5.7</u>	<u>62.0±5.1</u>	61.7±6.3
	RadCLIP	54.4±4.7	52.9±4.2	51.0±2.7	52.6±2.7	52.9±2.5
	VoCo	55.1±7.6	54.9±9.3	59.3±5.8	61.0±4.4	<u>62.1±3.9</u>
	Swin-UNETR	54.5±8.5	55.2±8.3	58.9±2.2	59.9±2.7	61.0±3.8
	MiM	56.0±7.8	54.8±10.8	56.9±7.3	55.7±4.4	57.5±5.6
	KAMP (ours)	62.5±8.7	63.6±4.4	65.0±5.3	67.6±3.9	70.2±1.8
FedBCa (2 classes)	DRIM	75.6±7.8	72.5±7.0	74.0±4.5	75.0±4.9	74.8±3.3
	HLIP	67.7±6.9	68.0±11.0	69.5±5.2	68.7±4.1	71.8±2.5
	Med3D	62.0±12.7	65.0±6.8	65.9±6.0	68.8±3.4	72.3±2.2
	VoCo	68.1±7.2	71.5±5.1	72.0±3.7	71.9±3.2	72.5±4.3
	RadCLIP	65.6±4.0	62.1±2.8	63.1±1.9	65.4±3.3	67.8±3.9
	Swin-UNETR	56.7±8.2	60.2±3.7	63.4±4.2	66.1±3.0	66.9±2.7
	MiM	71.1±7.6	<u>74.6±2.7</u>	<u>75.7±5.0</u>	<u>75.1±3.6</u>	76.1±2.9
	KAMP (ours)	<u>74.3±7.5</u>	76.7±4.4	78.8±5.5	78.4±3.8	82.6±3.6
TG-LIVT (2 classes)	DRIM	<u>57.2±10.2</u>	61.0±5.7	<u>62.9±3.1</u>	<u>65.7±2.5</u>	68.0±3.4
	HLIP	51.3±8.7	58.4±4.6	58.4±2.3	61.5±2.2	63.9±1.9
	Med3D	55.3±5.4	59.7±4.5	61.7±2.6	61.5±2.8	64.7±2.4
	VoCo	54.0±8.5	60.0±4.6	60.3±2.8	63.3±3.9	66.9±3.7
	RadCLIP	51.8±8.9	53.5±5.4	58.7±3.9	58.8±2.5	62.5±2.6
	Swin-UNETR	50.2±2.2	50.3±1.6	51.5±1.7	52.0±1.3	52.3±2.3
	MiM	53.9±1.5	57.7±2.8	59.4±1.4	62.7±3.3	63.6±2.7
	KAMP (ours)	59.4±4.8	64.8±5.62	67.6±1.5	71.7±2.7	73.9±2.5

the exact reduction is $\Delta H = H(h | z) - H(h | z, A) = I(h; A | z) \geq 0$ as in Eq. (5). Because the anchor is derived from metadata and clinical text and is visibility-aware and patient-specific, $I(h; A | z) > 0$ in practice. Consequently, the anchor prunes spurious cross-modal correlations, builds efficient links for semantic alignment, and yields more accurate and semantically grounded medical-image representations.

4. Experiments

4.1. Settings

Datasets. We pre-train KAMP on four public cohorts from The Cancer Genome Atlas (TCGA) [24], including TCGA-GBM, TCGA-LGG, TCGA-BLCA and TCGA-LIHC, covering brain, bladder and liver cancers. We use heterogeneous modalities comprising medical images (primarily MRI and CT), metadata of medical images, clinical text, pathology (H&E-stained whole slide tissue images), and genomics including bulk gene expression from RNA sequencing, mutations from whole genome sequencing, and copy number variations. TCGA-GBM and TCGA-LGG are brain tumor projects totaling 1,011 cases; TCGA-BLCA and

TCGA-LIHC contain 110 and 95 cases, respectively. Because only a small fraction of cases include medical imaging together with other modalities, we selected these TCGA datasets for their relatively extensive multimodal coverage. For the downstream task, we evaluate on two public datasets and one in-house dataset: BraTS23-MEN [2] for three-class meningioma grading with grades 1, 2, and 3; FedBCa [3] for binary bladder cancer subtype classification distinguishing NMIBC from MIBC; and the in-house TG-LIVT dataset for binary prediction of microvascular invasion (MVI).

Implementation Details. In Stage 1, we prompt GPT-5 [27] to bootstrap a textual knowledge base for the aligner; in Stage 2, we fine-tune Qwen3-8B [49] with reinforcement learning to generate texts; in Stage 3, we train the aligner using the Qwen3-8B-generated corpus. For the medical image branch, we initialize both the image encoder and the paired text encoder with RadCLIP [21], and the knowledge-aware aggregation block is implemented with a standard cross-attention module [43]. We apply low rank adaptation [13] to Qwen3-8B [49] and to the medical image encoder, while the text encoder remains frozen in both branches.

Evaluation metrics. We measure classification perfor-

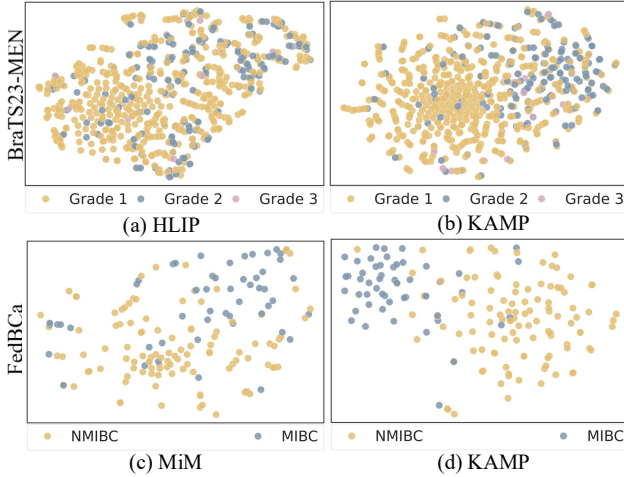


Figure 3. Clustering results of medical image features on BraTS23-MEN (top) and FedBCa (bottom).

mance with macro-AUC. To ensure robust estimates, we evaluate under the few-shot regime with $\{3, 5, 10, 15, 20\}$ shots by performing five independent runs. For each run, we randomly sample the required number of training cases per class without replacement, with no case overlap among training sets across runs. We report the mean and standard deviation of each metric over the five runs.

4.2. Few-Shot Performance Comparison

We evaluate the effectiveness of KAMP against single-modality baselines Swin-UNETR [40], VoCo [46, 47], MiM [57], dual-modality pretraining baselines RadCLIP [22], HLIP [53] and MRI-PTPCa [34], and the tri-modality method DRIM [31] under five few-shot settings (3, 5, 10, 15, 20) on three datasets, reporting macro-AUC (See Tab. 1).

BraTS23-MEN. On the three-class meningioma grading dataset BraTS23-MEN [2], KAMP achieves the best macro-AUC at every shot setting; at 5-shot setting it reaches 63.6%, surpassing the next best HLIP [53] by 5.5%, and at 20-shot setting it attains 70.2%, outperforming the second best VoCo [46, 47] by 8.1%. The advantage persists as supervision increases, indicating that the proposed paradigm surpasses single-modality, dual-modality, and tri-modality baselines while requiring a small amount of training data.

FedBCa. On the binary bladder cancer subtype dataset FedBCa [3], KAMP achieves the highest macro-AUC in most few-shot regimes. As the number of shots increases, KAMP consistently dominates, at 20-shot regime it reaches 82.6%, exceeding the next best MiM [57] by 6.5%, indicating stable scaling with data availability.

TG-LIVT. Across all few-shot regimes, our method achieves the highest macro-AUC and consistently surpasses

competing approaches by a clear margin. This advantage remains as the number of training data increases, and at the 20-shot setting, KAMP attains a state-of-the-art macro-AUC of (73.9%), exceeding the previous best model DRIM by 5.9%.

Figure 3 presents feature clusters obtained with t-SNE [42] on BraTS23-MEN [2] and FedBCa [3]. The arrangement of points shows that KAMP encodes high-level semantics in the feature space by amplifying medically meaningful image cues, yielding compact class-aligned clusters with enlarged inter-class separation, and thus achieving superior clustering quality compared with the strongest competing methods.

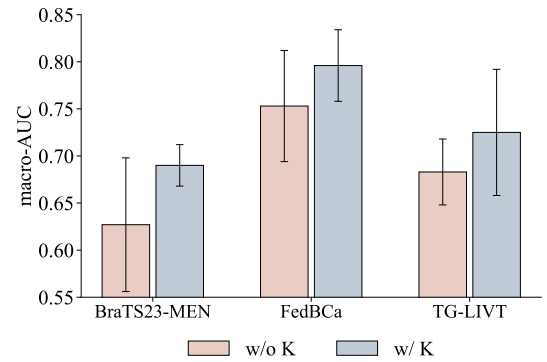


Figure 4. Ablation experiment on the BraTS23-MEN, FedBCa, and TG-LIVT dataset under 20-shot setting. “w/ K” uses knowledge generated by GPT-5 during alignment; “w/o K” aligns medical image embeddings directly to the fused pathology and genomics representation without any text. (Macro-AUC).

4.3. Ablation Studies

4.3.1. Effect of Knowledge Anchor

We quantify the contribution of textual knowledge in KAMP by pretraining two variants that differ only in whether teacher knowledge is injected into the cross-modal aligner, where the without-knowledge variant aligns medical image embeddings with the fused pathology and genomics representation without any text, and we keep all other components and the training protocol identical. Evaluation is conducted under the 20-shot setting and we report macro-AUC averaged over five independent runs with disjoint training sets. As shown in Figure 4, injecting textual knowledge yields consistent gains across all three datasets, improving macro-AUC by 6.3%, 4.3%, and 4.2% on BraTS23-MEN[2], FedBCa[3], and TG-LIVT, respectively. These results validate that textual knowledge strengthens alignment and translates into reliably better downstream performance. This ablation substantiates our core claim that LLM-derived textual knowledge functions as an effective semantic anchor that densifies super-

vision, prunes spurious cross-modal correlations, and reduces modality bias, which is central to the effectiveness of KAMP.

4.3.2. Effect of Reinforcement Learning for Knowledge Anchor

We assess how reinforcement learning improves the quality of the textual knowledge anchor and, in turn, the downstream performance of KAMP. As shown in Fig. 5, we compare four variants under the 20-shot setting on the FedBCa dataset: a non-RL baseline that directly uses GPT5-generated knowledge (w/ K) and three policy optimization strategies, PPO [32], DPO [29], and GRPO [35]. The non-RL baseline yields the lowest macro-AUC 79.6%. PPO [32], DPO [29] and GRPO [35] all improve over the non-RL baseline, suggesting that policy optimization helps calibrate the generated knowledge to better align medical image features with multimodal biomedical evidence. GRPO [35] achieves the highest macro-AUC 82.6%, surpassing all other variants with a clear margin. This confirms that GRPO [35] most effectively exploits the alignment signal as reward, promoting knowledge that is both patient-specific and semantically faithful to the imaging setting. In particular, GRPO-guided knowledge better emphasizes discriminative visual cues that are consistent with pathology and genomics, tightening alignment and translating into the strongest downstream accuracy. These results validate the necessity of reinforcement learning for KAMP and support the design choice of using GRPO [35] as the final policy optimization strategy for knowledge generation.

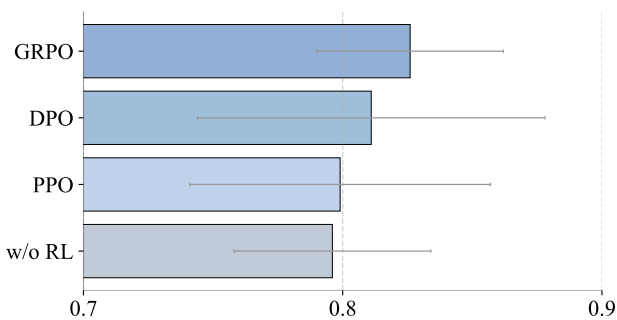


Figure 5. Performance comparison on FedBCa in the 20-shot setting (Macro-AUC). “w/o RL” denotes a LLM baseline using GPT-5 with no reinforcement learning; PPO, DPO, and GRPO are RL-tuned variants.

5. Conclusion

In this paper, we presented KAMP, a knowledge-anchored medical image pretraining framework that leverages LLM-derived personalized knowledge for cross-modal contrastive learning. Using generated text as a knowledge

anchor, KAMP provides fine-grained semantic cues and mitigates spurious correlations and modal bias, and produces more robust medical image representations. A three-stage training recipe first establishes an initial aligner with teacher knowledge, then optimizes the knowledge generator through GRPO[35] using the frozen aligner as rewards, and finally adapts the aligner under the learned knowledge distribution. Extensive experiments across multiple cancers and task families demonstrate consistent gains in few-shot regimes and SOTA results over single-modality, dual-modality and tri-modality baselines. Our results demonstrate that KAMP pushes the frontier of medical image pretraining in data scarce scenarios, converting patient context conditioned textual priors into precise and transferable guidance.

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